DeMarcus Crump

OPEN TO WORK

Artificial Intelligence, A.A.S. | Houston City College

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github.com/crumpdemarcus/Applied-AI-Portfolio

Portfolio Overview: A comprehensive collection of high-quality AI/ML projects across four course areas, showcasing proficiency in Deep Learning, Generative AI, Computer Vision, and NLP. Demonstrated ability to integrate LLMs (Llama 3.3, Gemini 2.5), build CNN visualizations, implement diffusion models, and deploy custom trained models for real-world utility.

12 PROJECTS | **4** COURSES | **15k+** LINES

01 // DEEP LEARNING

ITAI-2376

ConvoShrimp — CNN Viz

Interactive React app mapping CNN architecture to mantis shrimp biology. Animated walkthrough of convolution, pooling, and classification with embedded PyTorch code.

React 19 · Vite · TailwindCSS · Framer

Neural Network Squad — Education

Interactive web comic teaching weights, biases, and activation functions (ReLU, Sigmoid, Tanh) through illustrated characters and live slider demos.

HTML5 · CSS3 · JS · Zero Deps

Arrival — Seq2Seq

Encoder-decoder NLP experience exploring attention mechanisms and neural language representation inspired by the film.

React · Framer Motion · i18n

"Built CNN visualizations and diffusion model labs from scratch in PyTorch."

02 // AI APPLICATIONS

ITAI-2372

Aura Intelligence — Marketing GenAI

Dual-mode platform using Groq's Llama 3.3 70B. Storyteller Mode generates luxury copy; Curator Mode acts as a search-enabled assistant.

Flask · Groq API · TailwindCSS

AgentHub — Web3 + AI

Decentralized freelance marketplace. Uses AI for job matching and Smart Contracts for secure transaction settlement.

Python · Blockchain · Matching Algo

AeroSense — HealthTech

Respiratory health companion using Gemini 2.5 Pro for symptom interpretation and ML for environmental risk forecasting.

FastAPI · React · MongoDB · Gemini

03 // NLP

ITAI-2373

NuVision News — Semantic

Full-stack news analyzer featuring semantic clustering, bias detection, and knowledge graph generation.

React · TS · Hugging Face · Node.js

Text Preprocessing Lab

Engineered pipelines transforming raw text to features via TF-IDF, tokenization, stemming, and lemmatization.

NLTK · Pandas · Scikit-learn

Sentiment Analysis

Built binary and multi-class classification models. Focused on evaluation metrics and model tuning.

ML · Classification

Advanced NER

Named Entity Recognition using spaCy for extraction of complex entities from unstructured text.

spaCy · Python · Info Extraction

04 // COMPUTER VISION

ITAI-1378

Workplace Safety PPE — YOLOv8

Real-time object detection system for 10 classes of PPE. Custom trained YOLOv8 model with comprehensive metric evaluation.

PyTorch · OpenCV · Object Detection

CV Fundamentals

CNN architectures built from scratch alongside Transfer Learning implementations (VGG, ResNet) achieving 85%+ accuracy.

TensorFlow · Keras · CNNs

Advanced Architectures

Deep CNN optimization, hyperparameter tuning, and rigorous error analysis for iterative model improvement.

Fine-tuning · Optimization

"Leveraging vision transformers and CNNs for actionable insights."

TECH STACK: Llama 3.3 · Gemini 2.5 Pro · YOLOv8 · BERT/BART · Python · TypeScript · PyTorch · TensorFlow · Scikit-learn · Pandas · React · FastAPI · Flask · MongoDB · TailwindCSS · Git · Jupyter · Framer Motion

Repository Status: ● Public

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